

Homogeneous tails rule out strict adjoint cyclicity

Automated mathematical discovery run `fa_banach_001`

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Status. Candidate full solution, likely valid, subject to human review. For every regular unitarily invariant Hilbert function space in the source, neither the full multiplier algebra nor any of its norm-closed subalgebras has a strictly $*$ -cyclic vector. This settles the source's characterization question and removes hypothesis (i) from its Theorem 5.11.

1 Source question

The source is D. P. Blecher and R. Clouâtre, *Null projections and noncommutative function theory in operator algebras*, arXiv:2404.04788v2 [1]. Let $\mathcal{H}$ be a regular unitarily invariant Hilbert space of analytic functions on the ball $\mathbb{B}_d \subset \mathbb{C}^d$. Write $\mathcal{M}(\mathcal{H})$ for its multiplier algebra and $\mathcal{A}(\mathcal{H})$ for the multiplier-norm closure of the polynomial multipliers.

A vector $\xi \in \mathcal{H}$ is *strictly $*$ -cyclic* for an operator algebra $\mathfrak{A}$ if

$$\mathfrak{A}^*\xi = \{a^*\xi : a \in \mathfrak{A}\} = \mathcal{H}.$$

This is exact equality, not merely density. On source p. 24, immediately after Theorem 5.11, the authors state that they do not know a characterization of the spaces $\mathcal{H}$ for which $\mathcal{A}(\mathcal{H})$ has no such vector. They prove the sufficient condition $\mathcal{A}(\mathcal{H}) \subsetneq \mathcal{M}(\mathcal{H})$. The source passage and the downstream theorem are reproduced below.

$\mathcal{A}(\mathcal{H})^\sim q(1 - z_K) = \mathcal{I}(\mathcal{H})q(1 - z_K)$. Upon applying Ω , we find $\mathcal{A}(\mathcal{H})^\sim P = \mathcal{I}(\mathcal{H})P$ as subsets in $B(\mathcal{H})$. If $P \neq 0$, then there is a unit vector $\xi \in \mathcal{H}$ in the range of P ; let $E \in B(\mathcal{H})$ denote the corresponding rank-one projection. Since $E \leq P$, we have $\mathcal{A}(\mathcal{H})^*E = \mathcal{T}(\mathcal{H})E$. Let $\eta \in \mathcal{H}$ be an arbitrary vector. Choose a rank-one operator $t \in B(\mathcal{H})$ such that $t\xi = \eta$. Then, $t \in K(\mathcal{H}) \subset \mathcal{T}(\mathcal{H})$ so there is $a \in \mathcal{A}(\mathcal{H})$ with $a^*E = tE$. This implies that $a^*\xi = a^*E\xi = tE\xi = \eta$. In other words, $\mathcal{A}(\mathcal{H})^*\xi = \mathcal{H}$, so that ξ is strictly $*$ -cyclic for $\mathcal{A}(\mathcal{H})$. $\square$

We can now give a complement to Corollary 5.8.

Theorem 5.11. *Let $q \in \mathcal{T}(\mathcal{H})^{**}$ be a closed projection such that $q\mathcal{A}(\mathcal{H}) = q\mathcal{T}(\mathcal{H})$. Assume that*

- (i) $\mathcal{A}(\mathcal{H})$ admits no strictly $*$ -cyclic vector in $\mathcal{H}$, and
- (ii) there exists a non-empty $\mathcal{M}(\mathcal{H})$ -totally null set.

Then, q is $\mathcal{M}(\mathcal{H})$ -totally null, and hence is $\mathcal{A}(\mathcal{H})$ -null.

Proof. By Lemma 5.10, we see that $q \leq z_K$. Hence, by Lemma 5.4, there is a closed $\mathcal{A}(\mathcal{H})$ -interpolation subset $E \subset \mathbb{S}_d$ such that $\Theta(q) = \chi_E$. By virtue of Theorem 5.5, to show that q is $\mathcal{M}(\mathcal{H})$ -totally null, it is then equivalent to show that E is $\mathcal{M}(\mathcal{H})$ -totally null, which follows at once from Theorem 5.7. $\square$

We do not know of a characterization of those spaces $\mathcal{H}$ for which $\mathcal{A}(\mathcal{H})$ admits no strictly $*$ -cyclic vector. Nevertheless, we can give a simple sufficient condition: such vectors cannot exist provided that $\mathcal{A}(\mathcal{H})$ is properly contained in $\mathcal{M}(\mathcal{H})$ (which is the case in many classical examples). Indeed, this follows from the following general fact.

If $A \subset B(H)$ is a commutative subalgebra *properly* contained in its commutant A' , then A admits no strictly $*$ -cyclic vector. To see this, assume that $\xi \in H$ is strictly $*$ -cyclic for A , and choose $b \in A'$ with $b \notin A$. Then, there is $a \in A$ such that $a^*\xi = b^*\xi$. Using that A is commutative, this further implies $a^*c^*\xi = b^*c^*\xi$ for every $c \in A$, and in turn we must have $a^* = b^*$ or $a = b$, a contradiction.

Some additional partial results on the existence of strictly cyclic vector can be

The answer within the stated class is uniform: all of these spaces have the nonexistence property. In fact, even $\mathcal{M}(\mathcal{H})$ has no strictly $*$ -cyclic vector.

2 Proof intuition

Multipliers can only move Taylor mass upward in homogeneous degree. Thus every high-degree tail is invariant. A fixed vector ξ has arbitrarily small norm in sufficiently high tails.

If $\mathfrak{A}^*\xi$ were all of $\mathcal{H}$, the open mapping theorem would give a single constant C such that every unit vector could be written as $a^*\xi$ with $\|a\| \leq C$. Apply this to a unit vector already in a very high tail. Pairing the equation $a^*\xi = y$ with y forces the scalar 1 to be bounded by C times the tail norm of ξ , which tends to zero. This contradiction is independent of the size of the multiplier algebra and of any commutant calculation.

3 An invariant-filtration obstruction

Lemma 1 (Tail obstruction). *Let $\mathfrak{A} \subset B(\mathcal{K})$ be a norm-closed operator algebra. Suppose there are nonzero closed $\mathfrak{A}$ -invariant subspaces*

$$\mathcal{K}_1 \supset \mathcal{K}_2 \supset \cdots$$

whose orthogonal projections P_n converge strongly to zero. Then no $\xi \in \mathcal{K}$ is strictly $$ -cyclic for $\mathfrak{A}$.*

Proof. Suppose $\mathfrak{A}^*\xi = \mathcal{K}$. The bounded conjugate-linear map

$$Q_\xi : \mathfrak{A} \longrightarrow \mathcal{K}, \quad Q_\xi(a) = a^*\xi,$$

is surjective. The open mapping theorem, applied equivalently to the conjugate Banach space of $\mathfrak{A}$, gives $C > 0$ such that every unit vector $y \in \mathcal{K}$ has a preimage $a \in \mathfrak{A}$ satisfying

$$a^* \xi = y, \quad \|a\| \leq C. \quad (1)$$

Choose a unit vector $y_n \in \mathcal{K}_n$ and an a_n as in (1). Invariance gives $a_n y_n \in \mathcal{K}_n$, so

$$1 = |\langle a_n^* \xi, y_n \rangle| = |\langle P_n \xi, a_n y_n \rangle| \leq C \|P_n \xi\|.$$

But $P_n \xi \rightarrow 0$, a contradiction. $\square$

The strong convergence assumption can equivalently be replaced by $\bigcap_n \mathcal{K}_n = \{0\}$ for a decreasing sequence of closed subspaces.

4 Full answer for Hilbert function spaces

For the source's $\mathcal{H}$, unitary invariance gives the orthogonal homogeneous decomposition

$$\mathcal{H} = \bigoplus_{m=0}^{\infty} \mathcal{H}_m,$$

where $\mathcal{H}_m$ consists of homogeneous polynomials of degree m . Regularity ensures that every $\mathcal{H}_m$ is nonzero. Put

$$\mathcal{H}_{\geq n} = \bigoplus_{m \geq n} \mathcal{H}_m, \quad P_{\geq n} = P_{\mathcal{H}_{\geq n}}.$$

The projections $P_{\geq n}$ converge strongly to zero.

Theorem 2 (No strict adjoint cyclicity). *Let $\mathcal{H}$ be a regular unitarily invariant Hilbert space of analytic functions on $\mathbb{B}_d$. If $\mathfrak{A}$ is any norm-closed subalgebra of $\mathcal{M}(\mathcal{H})$, then*

$$\mathfrak{A}^* \xi \neq \mathcal{H} \quad (\xi \in \mathcal{H}).$$

*In particular neither $\mathcal{M}(\mathcal{H})$ nor $\mathcal{A}(\mathcal{H})$ admits a strictly *-cyclic vector.*

Proof. Every multiplier preserves every $\mathcal{H}_{\geq n}$. Indeed, if the Taylor series of f has no term below degree n and φ is analytic, then the Taylor series of φf has no term below degree n . Hence

$$M_{\varphi} \mathcal{H}_{\geq n} \subset \mathcal{H}_{\geq n} \quad (M_{\varphi} \in \mathcal{M}(\mathcal{H})).$$

The same is therefore true for $\mathfrak{A}$. Apply Lemma 1 with $\mathcal{K}_n = \mathcal{H}_{\geq n}$. $\square$

Thus the relation $\mathcal{A}(\mathcal{H}) \subsetneq \mathcal{M}(\mathcal{H})$ used in the source is not needed. Even in a hypothetical equality case, the common algebra would still have no strictly *-cyclic vector.

5 Strengthening of the source theorem

The source defines a Toeplitz C^* -algebra $\mathcal{T}(\mathcal{H})$ generated by $\mathcal{A}(\mathcal{H})$ and a notion of $\mathcal{M}(\mathcal{H})$ -total nullity for projections in $\mathcal{T}(\mathcal{H})^{**}$. Its Theorem 5.11 says that a closed projection q with $q\mathcal{A}(\mathcal{H}) = q\mathcal{T}(\mathcal{H})$ is $\mathcal{M}(\mathcal{H})$ -totally null, hence $\mathcal{A}(\mathcal{H})$ -null, provided that

1. $\mathcal{A}(\mathcal{H})$ has no strictly *-cyclic vector; and
2. a nonempty $\mathcal{M}(\mathcal{H})$ -totally null set exists.

Corollary 3 (Theorem 5.11 without hypothesis (i)). *Let $\mathcal{H}$ be as in Theorem 2, and let $q \in \mathcal{T}(\mathcal{H})^{**}$ be a closed projection satisfying*

$$q\mathcal{A}(\mathcal{H}) = q\mathcal{T}(\mathcal{H}).$$

If there exists a nonempty $\mathcal{M}(\mathcal{H})$ -totally null set, then q is $\mathcal{M}(\mathcal{H})$ -totally null and hence $\mathcal{A}(\mathcal{H})$ -null.

Proof. Theorem 2 verifies hypothesis (i) of source Theorem 5.11 automatically. Apply that theorem with its remaining hypothesis (ii). $\square$

No assertion is made when the source’s existence hypothesis (ii) fails.

6 Verification, limitations, and novelty audit

Eight checks were made against the proof. First, exact strict cyclicity was kept distinct from dense/topological cyclicity. Second, conjugate-linearity causes no open-mapping issue after passing to the conjugate Banach space. Third, high-degree tails are invariant under arbitrary analytic multipliers, not merely polynomial norm limits. Fourth, the homogeneous decomposition is exhaustive, so its tail projections converge strongly to zero. Fifth, regularity supplies nonzero test vectors in arbitrarily high degrees. Sixth, the proof covers the commutant-equality edge that the source’s sufficient condition misses. Seventh, no claim is made for truncated finite-dimensional function spaces, where arbitrarily high nonzero tails are absent. Eighth, Corollary 3 removes only source hypothesis (i) and preserves the independent total-null-set hypothesis.

The result also follows directly from the quotient-map formulation: strict $*$ -cyclicity would make Q_ξ a bounded surjection, whereas the unit vectors $y_n \in \mathcal{H}_{\geq n}$ prove that its adjoint cannot be bounded below. This supplies an independent functional-analytic reading of the same contradiction.

A bounded novelty search used the exact source title, arXiv id, authors, and the phrases “strictly $*$ -cyclic vector”, “strictly star-cyclic”, “multiplier algebra”, “Hilbert function space”, and “unitarily invariant”. The current arXiv v2 source, updated in September 2025, still contains the open statement. Located literature concerns forward strict cyclicity of weighted shifts or general strictly cyclic algebras, not the adjoint multiplier-algebra question. No explicit later answer or this homogeneous-tail argument was found. Novelty confidence is therefore provisional pending expert review.

References

- [1] D. P. Blecher and R. Clouâtre, *Null projections and noncommutative function theory in operator algebras*, arXiv:2404.04788v2 (2025), especially Theorem 5.11 and p. 24.